

Xin Wang

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Ph.D. Student in Transportation Engineering

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[Homepage](#) | [Google Scholar](#)

Research Interests

- Areas Transportation AI; Intelligent Transportation Systems; cybersecurity and reliability of data-driven transportation systems.
- Topics Robust traffic forecasting; traffic state estimation; adversarial attacks and defense; machine unlearning; optimization and sensitivity analysis; trustworthy AI for transportation operations.

Education

- 2022–Present **Ph.D., Transportation Engineering**, *University of Washington*, Seattle, WA
Advisor: Prof. Xuegang Jeff Ban
- 2020–2022 **M.S., Statistics**, *Renmin University of China*, Beijing, China
- 2016–2020 **B.S., Applied Mathematics**, *Central South University*, Changsha, China

Research & Teaching Experience

- 2022–Present **Research Assistant**, *University of Washington*, Seattle, WA
- Conduct research on cybersecurity, robustness, machine unlearning, and optimization for intelligent transportation systems.
 - Develop sensitivity-analysis and influence-function methods for evaluating how data perturbations affect transportation AI models and downstream decisions.
- Autumn 2024, **Teaching Assistant**, *CET 513: Optimization in Transportation*, UW, Seattle, WA
- Autumn 2025 ○ Led lab sections, held office hours, and graded assignments on optimization models and methods for transportation applications.

Industry Experience

- Jan–May 2021 **Machine Learning Engineer Intern**, *Baidu Inc.*, Beijing, China
- Worked on multi-objective learning-to-rank optimization for online video search.
 - Improved ranking quality and user engagement metrics using Pareto-efficient learning-to-rank and NSGA-II based solution search.

Selected Publications

Constrained machine unlearning of learning-based intelligent transportation systems models

Xin Wang, R. Tyrrell Rockafellar, Xuegang Jeff Ban

Transportation Research Part C: Emerging Technologies 190, 105763 (2026) .

Set-Valued Sensitivity Analysis of Deep Neural Networks

Xin Wang, Feilong Wang, Xuegang Jeff Ban.

Proceedings of the AAAI Conference on Artificial Intelligence (2025)

Transferability in Data Poisoning Attacks on Spatiotemporal Traffic Forecasting Models

Xin Wang, Feilong Wang, Yuan Hong, Xuegang Jeff Ban.

Transportation Research Part C: Emerging Technologies 183, 105501 (2026).

Data Poisoning Attacks on Traffic State Estimation and Prediction

Feilong Wang, Xin Wang, Yuan Hong, R. Tyrrell Rockafellar, Xuegang Jeff Ban.

Transportation Research Part C, 168, 104577 (2024).

Data Poisoning Attacks in Intelligent Transportation Systems: A Survey

Feilong Wang, Xin Wang, Xuegang Jeff Ban.

Transportation Research Part C, 165, 104750 (2024).

Model-Targeted Data Poisoning Attacks against ITS Applications with Provable Convergence

Xin Wang, Feilong Wang, Yuan Hong, R. Tyrrell Rockafellar, Xuegang Jeff Ban.

arXiv:2505.03966 (2025).

Infrastructure-Enabled Defense Methods against Data Poisoning Attacks on Traffic State Estimation and Prediction

Feilong Wang, Xin Wang, Xuegang Jeff Ban.

Conference in Emerging Technologies in Transportation Systems (TRC-30), 2025.

Invited Talks & Guest Lectures

Jan 2025 **Data Poisoning Attacks on Traffic State Estimation and Prediction**
ISTTT25

June 2024 **A Review of Data Poisoning Attacks in Intelligent Transportation Systems**
TRB 2025

Academic Service

Reviewer: *Transportation Research Part C*, TRB Annual Meeting, ICML, AAAI, NeurIPS

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